Analysis and Visualization of Earthquake Patterns (2021-2025) in the South Lebak Region Based on the Dataset from Gugus Mitigasi Lebak Selatan

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Abstract*—This study investigates earthquake patterns in South Lebak, Indonesia, using the official dataset provided by Gugus Mitigasi Lebak Selatan (GMLS) covering April 2021–April 2025. Three key parameters—magnitude, depth, and epicentral distance to Panggarangan Village—were analyzed after a preprocessing stage to ensure data completeness and consistency. Visual analyses included bar charts, two-dimensional and three-dimensional scatter plots, and geospatial overlays with tectonic plate boundaries. The results showed that shallow earthquakes (<70 km) with micro-to-small magnitudes (M 2–4) dominate the seismic profile and are concentrated within 160 km of the reference point. Intermediate- and deep-focus events were infrequent, while moderate-to-strong earthquakes occurred sporadically but remained critical owing to their potential impact on nearby communities. Clustering patterns broadly align with the southeastward motion of the Eurasian Plate, although the 2025 distributions appeared more scattered. A key contribution of this study is the integration of interactive visualizations into the GMLS platform, enabling local stakeholders and communities to directly explore seismic data in real time. This approach not only strengthens earthquake risk communication but also provides a replicable model for community-based disaster preparedness in tectonically active regions.

Keywords— Earthquake, Seismicity, South Lebak, Geospatial Analysis, Data Visualization

I. INTRODUCTION

Earthquakes are among the most destructive natural hazards globally, with severe implications for both human lives and infrastructure resilience. In addition to technological advances in disaster communication [1], understanding the dynamics of seismic events is crucial for improving disaster preparedness. Over the past two decades, Indonesia has experienced significant seismic activity owing to its tectonic position at the convergence of the Indo-Australian, Eurasian, Pacific, and Philippine Sea Plates [2]–[5]. The southern region of Java Island, where the Sunda megathrust accommodates oblique convergence, is particularly prone to shallow-to-intermediate earthquakes capable of generating both ground shaking and tsunami hazards [6], [7].

In the Banten Province, the South Lebak region is one of the most seismically active zones in West Java, located near the transition between the Sunda Strait and the southern Java subduction zone. The M6.1 Lebak earthquake on January 23, 2018, caused widespread ground shaking and highlighted the local community's vulnerability to seismic hazards [8]. While studies have identified active crustal fault systems such as Cimandiri and Lembang that exhibit shallow seismicity, comprehensive spatiotemporal analyses in South Lebak remain limited. Most existing research focuses on descriptive mapping of epicentral distribution and fault identification [9]–[11], without integrating analytical visualization or community-based interpretation that could enhance public awareness and preparedness.

To address this research gap, this study develops an analytical and visual framework for earthquake data interpretation using a locally sourced dataset from GMLS (Gugus Mitigasi Lebak Selatan), covering the period from April 2021 to April 2025. The dataset wad processed and visualized using Python, exploring the relationships among magnitude, depth, epicentral distance, and temporal patterns. This approach not only provides insights into regional seismic behavior but also introduces a data-driven model for translating scientific findings into interactive visualizations that support public disaster risk awareness.

Compared with previous local studies, this study introduces a distinctive integration of data analytics, geospatial visualization, and disaster communication, bridging the gap between earthquake science and disaster risk reduction practices. Furthermore, it aligns with the objectives of the Sendai Framework for Disaster Risk Reduction 2015–2030 by promoting risk understanding through open data and visualization [15], and supports Indonesia's National Agency for Disaster Countermeasure (BNPB) Regulation No. 4/2012 on community-based disaster preparedness [16]. The approach also reflects the analytical trends presented in Seismological Research Letters and Tectonophysics, where the integration of local datasets with global seismic frameworks is increasingly emphasized to enhance interpretability and applicability [17]. Thus, these findings contribute to advancing the scientific understanding of regional seismicity and strengthening local disaster mitigation initiatives.

II. RESEARCH METHODOLOGY

Building on the background described in the Introduction, this research methodology was designed to transform raw earthquake records into structured analyses and visualizations that could directly inform both scientific understanding and local disaster mitigation efforts. The process followed a sequential workflow, beginning with data acquisition and culminating in the integration of the results into the GMLS platform. The workflow is illustrated in Figure 1.

The workflow was divided into seven key stages: (A) Data Pre-processing, (B) Categorization of Earthquakes, (C) Distribution Analysis by Magnitude and Depth, (D) Pattern Analysis Using Bar Charts, (E) Pattern Analysis Using Scatter Plots, (F) Analysis of Earthquake Movements in Relation to

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Tectonic Plates, (G) Implementation of Visualization Results into the GMLS Website, and (H) Mitigation Implications.

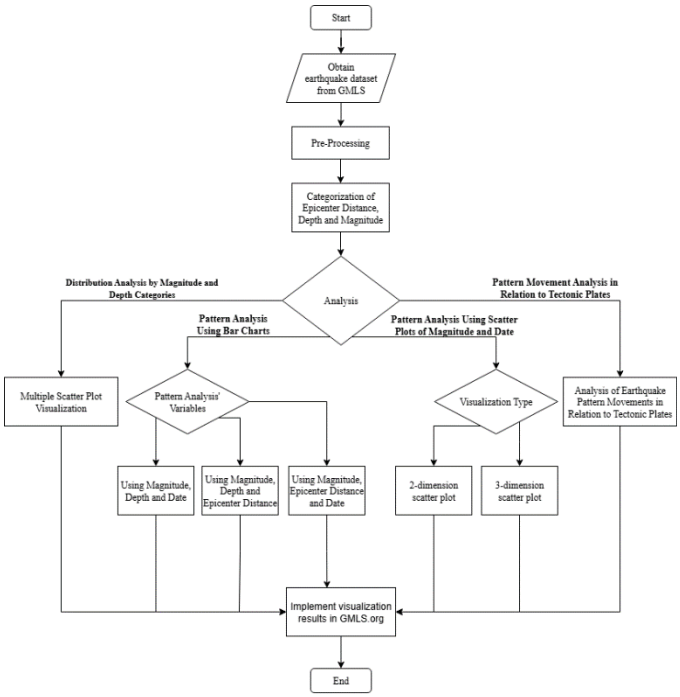


Fig. 1. Research Methodology Flowchart

A. Pre-processing

Earthquake records were obtained from the Gugus Mitigasi Lebak Selatan (GMLS), sourced from the Geophysical Station Class I – Tangerang under the Indonesian Agency for Meteorology, Climatology, and Geophysics (BMKG). The dataset, provided in the XLS format, includes the event date, epicentral coordinates (latitude and longitude), depth, and magnitude. The regional description field was excluded because location-specific socio-economic impact analysis was beyond the scope of this study.

The BMKG station functions as a primary seismic monitoring reference for western Java, ensuring that the dataset represents officially validated earthquake records [18]. The dataset structure—comprising the event date, epicentral coordinates, depth, and magnitude—followed the same fundamental metadata parameters used in the USGS Earthquake Catalog [19]. This consistency ensures interoperability with international seismic data formats and facilitates comparative analysis in future research.

Each entry was inspected for missing values across all the features. No missing data were found, resulting in a complete dataset of 1,835 earthquake events recorded between April 2021 and April 2025. This pre-processing step ensured dataset integrity and reliability for subsequent categorization and visual analysis.

B. Categorization of Earthquakes

After preprocessing, earthquakes were categorized according to three main parameters:

- The magnitude classification followed the BMKG standard [20], which defines micro (<3), small (3–4), moderate (4–5), and destructive (>5) earthquakes, consistent with international seismological conventions.
- Depth was grouped into shallow (<70 km), intermediate (70–300 km), and deep (>300 km), following the BMKG and international standards [20], [21]. Shallow earthquakes are typically associated with crustal faulting and tend to cause greater surface damage, whereas intermediate and deep events are linked to subduction-related processes [5], [17]
- Epicentral Distance – measured from Panggarangan Village, the site of the GMLS Command Center with latitude -6.910938335139458 and longitude 106.21900678001691. The coordinates were obtained using Google Maps [22]. Distances were calculated using the Haversine formula [23], which is more accurate than the Euclidean method for geodesic measurement. Following the attenuation principles of seismic intensity [24], that every 100-mile increase in distance (approximately 160.9 km) reduces earthquake intensity by four-thirds of the intensity near the epicenter. Distances were divided into four categories: <80 km, 80–160 km, 160–240 km, and >240 km. This division assumes that each 80 km increment potentially decreases earthquake intensity by roughly half of Neumann's reduction, allowing the distance categories to better represent the differences in earthquake impact risk for the Panggarangan area.

This standardized classification provides a consistent framework for analyzing spatial and temporal earthquake distributions, forming the basis for subsequent mapping and visualization processes.

C. Distribution Analysis by Magnitude and Depth Categories

A matrix-based approach was applied to map the earthquake distributions. Rows represent depth categories (shallow, intermediate, and deep) and columns represented magnitude categories (ultra-micro, small, moderate, strong, and major). Each cell in the matrix displays the number and spatial distribution of earthquake epicenters within the corresponding category pair.

This approach enabled a structured identification of dominant seismic patterns in South Lebak, particularly in distinguishing the prevalence of shallow and low-magnitude earthquakes from intermediate and deep events. Such categorical visualization helps to clarify the spatial concentration of seismicity, a method often employed in regional seismotectonic assessments [5], [17].

D. Analysis Using Bar Charts

Three stacked bar chart configurations were constructed:

- Magnitude–Year–Distance,
- Magnitude–Depth–Distance,
- Magnitude–Depth–Year.

These visualizations were designed to highlight annual variations in seismic activity, interactions between depth and epicentral distance, and the evolution of magnitude distributions over time. This method follows the common exploratory approach in regional seismic analyses, where

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stacked charts provide a clear depiction of the temporal and spatial variability in earthquake occurrence [5], [9], [17].

E. Pattern Analysis Using Scatter Plots

Scatter plots were developed in two forms:

- 2D Scatter Plots, with event date (x-axis) vs. magnitude (y-axis), depth encoded by color, and epicentral distance represented by point size. This representation was chosen to facilitate the observation of temporal trends in earthquake events, distribution of magnitudes across depths, and potential recurrence of events within certain time intervals.
- 3D Scatter Plots, with event date (x-axis), epicentral distance (y-axis), and magnitude (z-axis), while depth remained as color coding. This 3D approach was selected to provide a more comprehensive view of the data distribution, allowing for the identification of clustering tendencies based on a combination of time, distance, and magnitude.

By utilizing both visualization forms, this study aims to obtain a more complete depiction of seismic activity patterns, particularly regarding the distribution of magnitudes over time, variations in depth, and concentration of epicenter distances relative to the study area. This method not only facilitates the detection of patterns such as recurring shallow events or seasonal clustering but also provides visual evidence of the tectonic control of seismicity, consistent with previously reported crustal faulting mechanisms and oblique convergence patterns in western Java [4], [5], [17].

F. Analysis of Earthquake Movements in Relation to Tectonic Plates

At this stage, the study analyzed earthquake pattern movements in relation to tectonic plate dynamics by utilizing scatter plots overlaid on the map of Indonesia. Each earthquake event was color-coded according to the year of occurrence, enabling the identification of temporal changes in the distribution. Tectonic plate boundaries were added as spatial references, with data obtained from online repositories [21].

This visualization was designed to evaluate the relationship between earthquake locations and the direction of movement of the Eurasian Plate. By comparing the position and distribution of earthquake events across years, spatial displacement patterns can be analyzed to identify the dominant direction of seismic activity associated with plate dynamics. The analysis aimed to assess whether earthquake clusters align with plate movement trajectories and evaluate the consistency of these patterns throughout the observation period. Events were color-coded by year to highlight shifts in the distribution.

This approach allows researchers to visualize the temporal–spatial relationship between earthquakes and plate motion, providing a basis for understanding the tectonic mechanisms influencing seismic patterns in the study area [11,25,26]. The results of this analysis can serve as valuable input for disaster mitigation strategies that account for the directional movement of active tectonic zones.

G. Implementation of Visualization Results into the GMLS Website

All outputs were implemented using the GMLS online platform. Interactive results were exported in HTML format, whereas static outputs (e.g., magnitude–depth distribution maps) were stored as PNG files and integrated via PHP. This combination of static and interactive visualizations allows stakeholders and communities to directly access and explore seismic information, thereby supporting practical earthquake risk communication.

All data processing and visualization in this study were performed using Python. This ensured reproducibility, transparency, and efficiency throughout the workflow, while also enabling seamless integration of outputs into the GMLS platform.

H. Linkage to Mitigation Implications

Although the primary focus of this study lies in the processing, categorization, and visualization of earthquake data, its broader objective extends to strengthening disaster preparedness and community resilience. Visualization design follows a human-centered communication approach. This approach aligns with the Sendai Framework for Disaster Risk Reduction 2015–2030 [14], which emphasizes the critical role of accessible risk information and inclusive multi-stakeholder participation in reducing disaster impacts.

Beyond its technical function, the integration of research visualizations into the GMLS platform represents a novel contribution to local disaster preparedness. By connecting scientific outputs to practical dissemination channels, this implementation directly supports the BNPB's regulation of community-based disaster management [15], ensuring that hazard information is not confined to experts but is actively shared with local stakeholders.

III. Results and Discussion

Building on the methodological framework described in the previous section, this study begins the analysis with the pre-processing stage, which ensures that the dataset is clean and suitable for subsequent visualization and interpretation.

A. Pre-processing

The dataset analyzed in this study comprises 1,835 earthquake events. Each record includes event date, epicentral coordinates (latitude and longitude), depth, and magnitude. Table I summarizes the structure and completeness of the dataset.

TABLE I. Dataset Structure of Earthquakes

Date	***Lat***	***Lng***	***Depth***	***Mag***	***Region***
01/04/2021	-6,46	104,89	10	3,5	Sunda Strait, Indonesia
01/04/2021	-7,75	106,39	3	2,9	Java, Indonesia
07/04/2021	-7,30	104,88	10	3,6	Southwest of Sumatra, Indonesia
...	...	...	...	...	...
30/04/2025	-8,36	106,02	10	2,9	South of Java, Indonesia

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Completeness verification confirmed that no missing or inconsistent entries were present, resulting in a clean dataset suitable for direct analysis without further correction. The regional description field, although present in the original data, was excluded because this study focused exclusively on the physical and tectonic characteristics of seismicity rather than socio-economic impacts. Thus, the final dataset thus represents a fully validated and standardized record of seismic activity in western Java, ensuring reliability for subsequent classification and visualization analyses.

B. *Categorization of Earthquakes*

After preprocessing, the dataset of 1,835 events was organized into categorical groups based on magnitude, depth, and epicentral distance from Panggarangan Village. This step applied the classification schemes outlined in the methodology but focused on the outcomes of their application to the dataset.

In general, this categorization demonstrates that South Lebak's seismicity is characterized by frequent, low-magnitude, shallow-focus earthquakes concentrated near the local tectonic boundary. These structured categories provide an analytical foundation for spatial and temporal pattern analyses in the subsequent sections.

C. *Distribution Analysis by Magnitude and Depth Categories*

The analysis (Figure 2) shows that seismicity in South Lebak is dominated by shallow earthquakes (<70 km), mostly within the micro (M < 3), small (3–4), and moderate (4–5) ranges. These shallow events account for the majority of the 1,835 records, indicating that near-surface faulting is the primary source of local seismic activity [5], [17]. In contrast, intermediate (70–300 km) and deep (>300 km) earthquakes are rare, which is consistent with the regional subduction characteristics along the Java margin [4], [17].

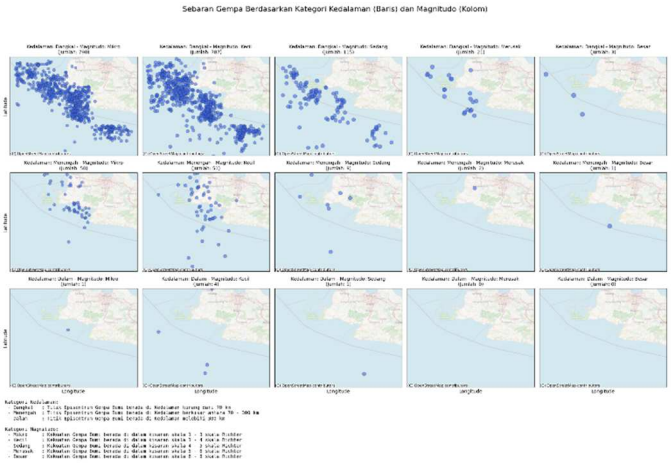


Fig. 2. Earthquake Distribution by Depth and Magnitude Categories

Only a few strong (M 5–6) events were detected, and no destructive deep-focus earthquakes occurred during 2021–2025. This pattern aligns with findings from West Java, where shallow, low-to-moderate magnitude events dominate near active faults such as Cimandiri and Lembang [9], [10]. Similar trends hsve been reported in West Sumatra and other Indonesian regions [27]–[30].

Comparative studies have supported this interpretation. Salsabillah and Prastowo [28] highlighted that shallow seismic sources tend to produce smaller but more frequent earthquakes, whereas intermediate to deep sources, though less common, are capable of generating larger and potentially damaging events. Basid et al. [30] further showed that many destructive earthquakes in mainland West Java were associated with shallow active faults. These findings align with the South Lebak case, where shallow seismicity is dominant but occasional moderate events underscore the importance of sustained monitoring and hazard preparedness.

D. *Pattern Analysis Using Bar Charts*

The bar chart analysis provided additional insights into the temporal and spatial distribution of seismic events. Figure 3 illustrates the distribution of earthquakes by depth, magnitude, and year. The results confirmed the dominance of shallow earthquakes across all magnitude categories, with peak activity observed between 2022 and 2024. These shallow clusters correspond to active crustal deformation zones influenced by the Java subduction system and local fault networks [4], [5], [17].

In contrast, large magnitude events were rare, recorded only at shallow and intermediate depths, and were absent in 2025. Intermediate and deep focus events were sporadic and lacked temporal consistency, aligning with observations that such earthquakes originateed from subducting slab interactions beneath Java [17]. Large-magnitude events (M > 5) were confined to shallow or intermediate depths, confirming that the hazard potential increases when moderate earthquakes occur near the surface [9], [10].

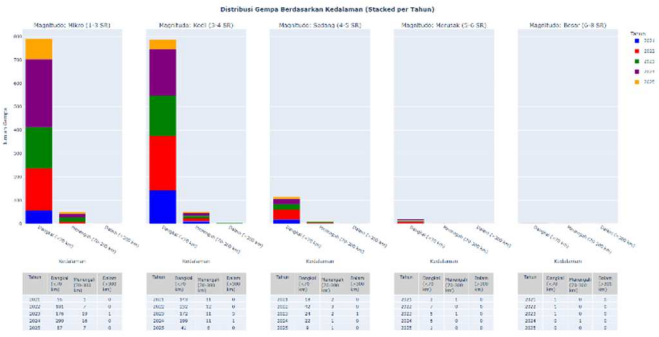


Fig. 3. Distribution of Earthquakes by Depth, Magnitude, and Year.

Figure 4 shows the distribution by depth category and epicentral distance. The visualization indicates that shallow earthquakes dominate across nearly all magnitude classes, particularly within 160 km of the reference point. Intermediate-depth earthquakes occurred in smaller numbers, whereas deep-focus events are extremely rare. Even large-magnitude events remained concentrated at shallow depths at relatively short epicentral distances.

A different trend emerges in Figure 5, which displays the distributions by epicentral distance, magnitude, and year. For the micro and small categories, seismic activity fluctuated with an increasing trend until 2024, followed by a sharp decline in 2025. Most of these events occurred within 160 km, whereas earthquakes beyond 240 km were almost absent. For moderate to large magnitudes, the number of events was significantly smaller, yet their distribution remained concentrated at short distances.

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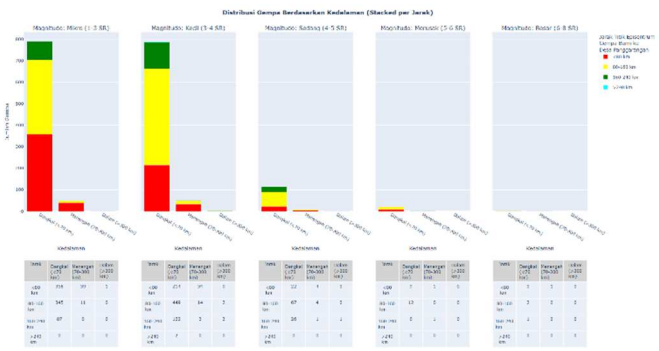


Fig. 4. Distribution of Earthquakes by Depth, Magnitude, and Epicenter Distance.

The three bar chart visualizations consistently confirm that South Lebak's seismicity is dominated by shallow, micro-to-small magnitude events clustered within 160 km of the study area. Although moderate-to-large earthquakes are infrequent, their potential impacts remain critical, especially when they occur near populated areas [27], [28]. These findings emphasize the need for disaster mitigation strategies that prioritize monitoring and early warning for shallow earthquakes, with particular attention paid to moderate events that could trigger significant local damage despite their lower frequency.

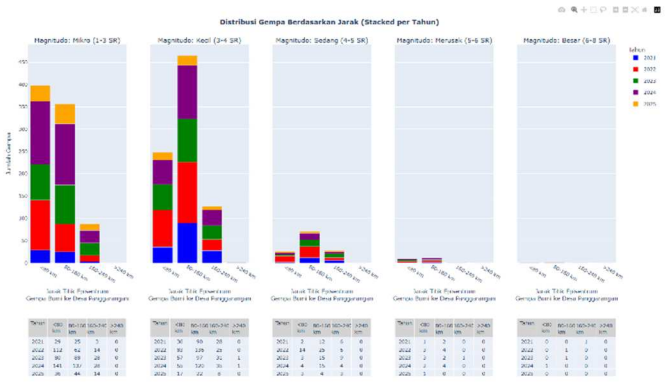


Fig. 5. Distribution of Earthquakes by Epicenter Distance, Magnitude, and Year.

E. *Pattern Analysis Using Scatter Plots*

Scatter plot visualizations provide a complementary perspective on the spatio–temporal patterns of seismicity in South Lebak.

Figure 6 shows the 2D scatter plot of magnitude versus time, with depth and distance encoded. The visualizations emphasize the temporal variability and clustering behavior of seismic events in South Lebak. Unlike the bar chart analysis that highlights the frequency distribution, these plots reveal episodic sequences of shallow earthquakes (M 2–M 4) recurring within short time intervals, particularly during 2021, 2022, and late 2024. Such short-term clustering suggests periods of intensified local stress accumulation along active faults in the forearc zone [4], [17]. A notable sequence was identified in May 2023, when a series of earthquakes ranging from M2 to M5 occurred over consecutive days, most of which were located 80–160 km from the reference point.

Intermediate-depth earthquakes (70–300 km) appeared less frequently, generally with magnitudes of M3–M4, but included one significant event exceeding M6 in January 2024. Deep-focus events (>300 km) were rare, with only six recorded during 2023–2024, all in the M3–M4 range and without extreme variations. This indicates that while deep seismicity is minimal in South Lebak, intermediate sources still hold potential for strong shaking [10].

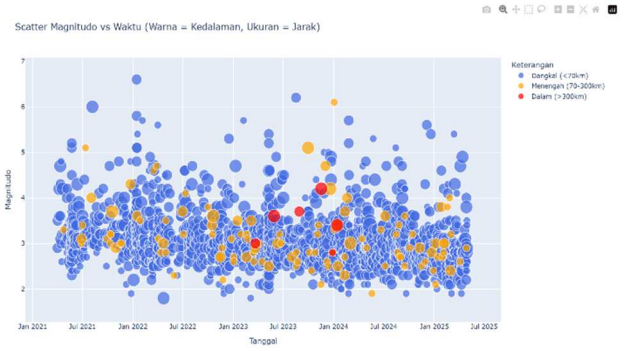


Fig. 6. 2D Scatter Plot of Magnitude, Distance, and Year

Figure 7 presents a 3D scatter plot, which highlights the clustering of events in specific periods, particularly April 2021, March 2022, and late 2024 to early 2025. These clusters were dominated by shallow earthquakes of small to moderate magnitudes (around M3), concentrated within 100 km of Panggarangan Village. Intermediate- and deep-focus events appeared sporadically and showed no consistent recurrence patterns.

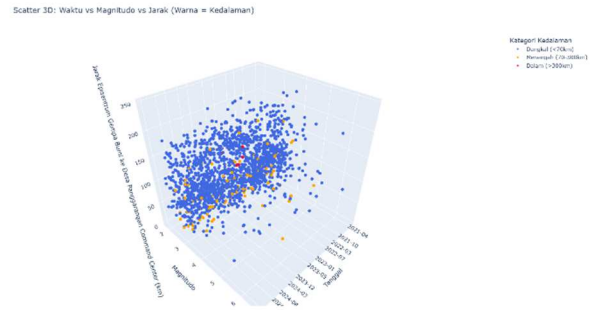


Fig. 7. 3D Scatter Plot of Time, Magnitude, and Distance

Overall, scatter-based temporal mapping indicates that South Lebak's seismicity occurs in episodic bursts rather than uniform patterns, dominated by shallow swarms interspersed with isolated moderate shocks — consistent with regional stress release cycles [17]. These results emphasize the importance of continuous monitoring, as shallow clustering and occasional moderate-to-strong events could pose risks to the surrounding communities.

To provide a clearer overview, Table II summarizes the scatter plot findings by depth category, highlighting the frequency, magnitude ranges, and notable events identified during the observation period. Shallow earthquakes dominate both the frequency and magnitude range, with distinct clusters in 2021, 2022, and late 2024. Intermediate events, although sporadic, include a significant case in January 2024 (>M6). Deep focus earthquakes are rare and do not exhibit significant clustering. These findings reinforce that South Lebak's seismicity is primarily shallow in origin, but occasional

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moderate-to-strong events still warrant continuous monitoring and preparedness [10],[27],[28].

F. *Analysis of Earthquake Pattern Movements in Relation to Tectonic Plates*

The spatial distribution analysis (Figure 8) reveals that most earthquake events in South Lebak are aligned with the subduction boundary south of Java Island, reflecting the oblique convergence between the Indo-Australian and Eurasian plates. This alignment confirms that the regional seismicity is largely controlled by subduction-related mechanisms rather than by intraplate deformation. Similar patterns have been described in previous studies across Java, where oblique subduction drives both megathrust activity and crustal faulting along the southern margin of the island [4], [5], [6], [17].

TABLE II. SUMMARY OF SCATTER PLOT FINDINGS BY DEPTH CATEGORY

Depth Category	***Magnitude Range Observed***	***Frequency & Pattern***	***Notable Events***	***Spatial Concentration***
Shallow (<70 km)	M2–M6+	Dominant and continuous throughout observation period	May 2023 sequence (M2–M5, multiple days)	Mostly <100 km; peak clusters 2021, 2022, late 2024
Intermediate (70–300 km)	M3–M6+	Sporadic, without annual consistency	Jan 2024 event >M6	Mainly within 80–240 km
Deep (>300 km)	M3–M4	Very rare (only six events, 2023–2024)	None significant	No clear clustering

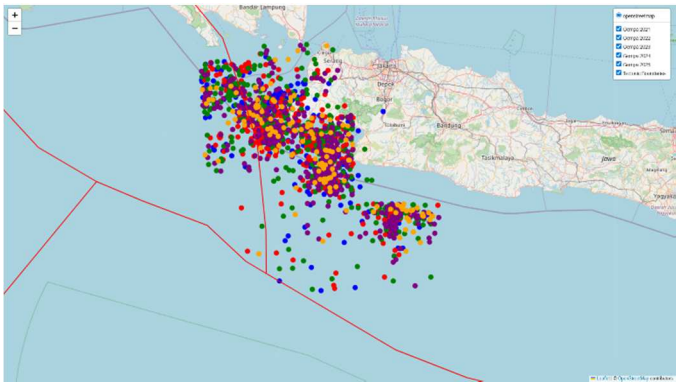


Fig. 8. Earthquake Distribution on the Map of Indonesia, 2021–2025

The results show that shallow seismicity (<70 km) dominates the dataset, consistent with the tectonic setting in which the descending Indo-Australian slab interacts with the overriding Eurasian crust. Such shallow events are typically associated with crustal faulting and intraslab deformation near the interface zone. This dominance of shallow events is in line with the national seismicity record reported by the BMKG [18], indicating that approximately three-quarters of earthquakes in western Java occur within 70 km of the surface.

Temporal color-coding of events from 2021 to 2025 illustrates gradual shifts in cluster locations—moving southeastward from Ujung Kulon toward the Panaitan and Panggarangan regions. This spatial transition is consistent with the general southeastward motion of the Eurasian Plate relative to the Indo-Australian Plate [4], suggesting that local stress accumulation and release occur progressively along the subduction interface. Comparable observations in West Java indicate that earthquake clusters often migrate along local fault systems such as the Cimandiri, Lembang, and Baribis faults [9], [10], emphasizing the interconnectedness of regional fault activity.

When compared with national datasets, such as the BMKG's seismic catalogue and the findings of Patria and Aulia [6], the observed distribution supports the interpretation that the South Lebak represents part of a broader seismotectonic corridor linking the Sunda Strait to the West Java fault system. This correspondence strengthens the argument that the seismic patterns f the area are not random but governed by regional tectonic controls.

Altogether, the combination of shallow event dominance, spatial clustering along the southern margin, and progressive southeastward migration patterns underscores the dynamic interaction between the subduction-driven stress accumulation and local crustal deformation. Understanding these relationships provides a geophysical basis for targeted disaster mitigation planning in South Lebak, particularly in areas directly above active subduction zones.

G. *Implementation of Visualization Results into the GMLS Website*

The integration of the visualization results into the GMLS website represents a practical application of the analytical workflow, allowing users to access earthquake information through an intuitive web-based platform (Figure 9). Within the Infographics menu, three main modules were provided: Dashboard, Geospatial Analysis, and Earthquake Analysis.

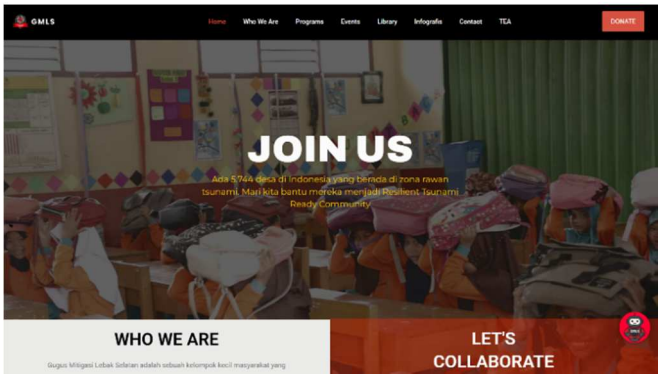


Fig. 9. GMLS Website Homepage

All visualization outputs—both static and interactive—are embedded in the Earthquake Analysis module (Figure 10). Through this module, users can explore earthquake distributions base on magnitude, depth, epicentral distance, and tectonic plate associations. Interactive plots were exported into the HTML format, enabling direct exploration without the need to rerun analysis scripts, while static visualizations (e.g., distribution matrices by magnitude and depth) were provided as PNG files integrated via PHP.

This implementation ensures that complex seismic data are presented in an accessible format, thereby enhancing the usability of multiple stakeholders. In particular, the platform supports monitoring activities, provides a pedagogical resource for earthquake education, and serves as a decision-support tool for local risk mitigation strategies. By combining technical accuracy with user-oriented design, the GMLS

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website demonstrates how community-level seismic data initiatives can bridge the gap between scientific analyses and practical disaster preparedness.

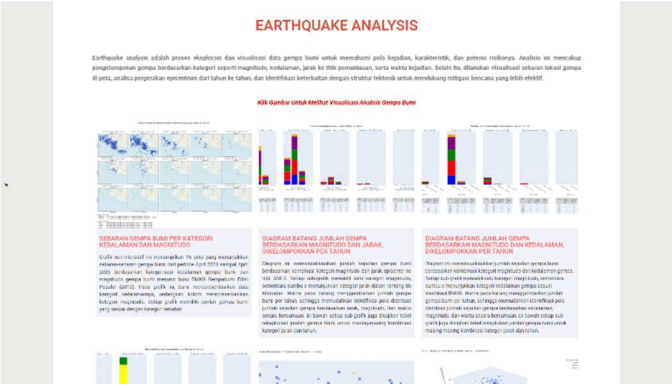


Fig. 10. GMLS Website Infographics Page

H. Mitigation Implications

This implementation follows a human-centered communication design, emphasizing clarity and usability to make scientific information more accessible to both experts and non-specialist audiences. The visualization interface simplifies the complex relationships among earthquake magnitude, depth, distance, and time, helping users intuitively recognize seismic patterns and potential risk zones. This approach aligns with the Sendai Framework for Disaster Risk Reduction 2015–2030 [14], which calls for the dissemination of open, understandable, and actionable risk information to enhance community resilience.

Beyond its technical deployment, GMLS integration demonstrates a novel contribution of this research by combining local-scale seismic datasets with digital visualization techniques to form an interactive decision-support system. This effort complements the BNPB regulation of community-based disaster management [15], by transforming scientific findings into participatory, data-driven knowledge tools for local preparedness and education. The resulting platform thus serves as both a repository and communication medium, ensuring that scientific insights are accessible for real-world mitigation efforts.

Experience from other regions illustrate that effective earthquake mitigation strategies require both technological readiness and community engagement. For example, in Bali, the development of on-site earthquake early warning systems was proposed as a way to deliver rapid alerts directly to vulnerable populations [31]. Such approaches emphasize the importance of decentralized solutions that can complement national-scale monitoring networks.

Lessons from the 2018 Palu–Donggala disaster further demonstrate that the effectiveness of early warning systems depends not only on technical capacity but also on public understanding and trust [32]. Despite the existence of official alerts, many communities were unable to respond effectively because of gaps in communication and preparedness. This highlights the critical role of education, regular drills, and community-based preparedness initiatives in bridging the "last mile" of warning dissemination.

The integration of interactive visualization into GMLS provides a concrete pathway for bridging the gap between scientific research and public preparedness. By enabling users to directly explore spatial and temporal earthquake patterns, the platform supports local monitoring, risk communication, and participatory learning. To maximize the impact, this study recommends adopting a dual-track strategy for mitigation.

- Technical reinforcement, through continuous monitoring and integration of new earthquake records into GMLS, ensures that visual analytics remain current and relevant.
- Community engagement, using GMLS visualizations as educational tools in local training, awareness programs, and school-based preparedness curricula.

IV. Conclusions

This study analyzed earthquake activity in South Lebak between April 2021 and April 2025 using categorized, visual, and geospatial approaches integrated through the GMLS platform. The findings confirm that the seismicity of the region is dominated by shallow, low-to-moderate magnitude events concentrated within 160 km of the Panggarangan area. Although major earthquakes are rare, their potential impact, particularly when they occur at shallow depths near active faults, remains a key concern for local resilience planning.

The main contribution of this study lies in the integration of localized seismic datasets with interactive visualization tools, creating a framework that connects scientific earthquake analysis to practical, community-level disaster mitigation. By transforming raw BMKG seismic data into accessible visual formats, the GMLS platform facilitates risk communication, public education, and data-informed decision-making. This represents a novel link between seismological research and community-based resilience building, addressing a critical gap in Indonesia's local-scale earthquake studies that have traditionally focused on technical analysis without mechanisms for public engagement.

From a methodological standpoint, this study demonstrates how open-data visualization can serve as a bridge between science and society, aligning with the Sendai Framework's emphasis on accessible risk information and participatory disaster reduction. The use of Python-based visualization and HTML integration ensures transparency, reproducibility, and adaptability for future updates.

However, this study had several limitations. The dataset relied primarily on historical records from BMKG's Geophysical Station Class I – Tangerang, which does not provide real-time updates or ground-motion parameters such as intensity or acceleration. Additionally, the temporal window (2021–2025) represents only a short segment of the regional seismic cycle, thus long-term trends cannot be inferred.

Future research should therefore aim to:

- Integrate real-time data streaming and early-warning modules into the visualization system.
- Combine seismic data with socio-economic and infrastructure vulnerability layers; and
- Conduct comparative validation using complementary datasets from the USGS or regional monitoring networks.

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In conclusion, this study established a replicable, data-driven visualization model for local seismic monitoring and education. this highlights that earthquake resilience in South Lebak—and potentially in other Indonesian regions—can be strengthened not only through advanced instrumentation, but also through accessible, participatory knowledge systems that empower communities to act before the next earthquake strikes.

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